

# **SIMATS ENGINEERING ASSESSMENT TOOL 1**

**COURSE CODE:** CSA0713  
**NETWORKS**

**COURSE NAME: COMPUTER**

## **COURSE OUTCOME COVERED**

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

*Assessment Tool Weightage: 40%*

Annexure A

## **ANALYTICAL PROBLEM SOLVING**

### ***Code of Conduct:***

*I NS BHAVESH/192565086 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

*NS BHAVESH*

***Signature of the student:***

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## Annexure A

### ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial  $X^3+1$ .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the

given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

**5.** A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps)**, **home Wi-Fi (20 Mbps)**, and **mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4<br>(Exemplary)                                                            | Level 3<br>(Proficient)                   | Level 2<br>(Developing)                     | Level 1<br>(Beginning)                        |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------|-----------------------------------------------|
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding ; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method   | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors         | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer     | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation           | No interpretation                             |

# ANSWER SHEET

Name / Reg No: \_\_\_\_\_

Course Code: CSA0713 | Course: Computer Networks | Assessment: AT1 (CO2)

## Question 1

Message = 11100011, Generator polynomial  $G(x) = X^3 + 1 \rightarrow G = 1001$  (degree  $r = 3$ )

### (i) Polynomial long division to find the transmitted codeword

Step 1: Append  $r = 3$  zero bits to the message (since the generator has degree 3).

Augmented message = 11100011 000 = 11100011000

Step 2: Divide the augmented message by the generator 1001 using modulo-2 (XOR) division.

11100011000 divided by 1001

1110 XOR 1001 = 0111

1110 XOR 1001 = 0111 (bring down next bit)

1110 XOR 1001 = 0111 (bring down next bit)

1111 XOR 1001 = 0110 (bring down next bit)

1101 XOR 1001 = 0100 (bring down next bit)

1000 XOR 1001 = 0001 (bring down next bit)

0010 (leading bit 0, no XOR, bring down next bit)

0100 (leading bit 0, no XOR - all bits consumed)

**Remainder (CRC / FCS) = last 3 bits = 100**

Step 3: Append the remainder to the original message to form the transmitted codeword.

Transmitted message = 11100011 + 100 = 11100011100

Check: Dividing 11100011100 by 1001 gives remainder 000, confirming the codeword is exactly divisible by the generator.

### (ii) Effect of inverting the leftmost bit

Original transmitted codeword = 11100011100

Leftmost bit (1) is inverted (1 to 0) due to noise, so the received codeword becomes:

Received = 01100011100

Dividing the received codeword by the generator 1001 (modulo-2 division):

01100011100 divided by 1001  $\rightarrow$  remainder = 010 (non-zero)

**Result: The receiver's CRC calculation produces a non-zero remainder (010).**

How the receiver detects the error: At the receiver, the received bit stream is divided by the same generator polynomial (1001). If the remainder is 000, the frame is assumed to be error-free. Since the remainder here is 010 (not all zeros), the receiver concludes that the frame was corrupted during transmission and requests retransmission or discards the frame.

## Question 2

### a) Throughput calculation

Given: Bandwidth = 10 Mbps, Frames = 12,000 per minute, Frame size = 10,000 bits/frame

Total bits transmitted per minute =  $12,000 \times 10,000 = 120,000,000$  bits/minute

Throughput =  $120,000,000 \text{ bits} / 60 \text{ seconds} = 2,000,000 \text{ bits/second}$

**Throughput = 2,000,000 bps = 2 Mbps**

(Note: the 10 Mbps figure is the bandwidth/capacity of the link, while the actual throughput - the data that is successfully passed - is only 2 Mbps.)

### b) Channel capacity calculation (Shannon's theorem)

Given: Bandwidth  $B = 3000 \text{ Hz}$ , Signal-to-Noise Ratio  $\text{SNR} = 3162$

$$C = B \times \log_2(1 + \text{SNR})$$

$$C = 3000 \times \log_2(1 + 3162)$$

$$C = 3000 \times \log_2(3163)$$

$\log_2(3163)$  is approximately 11.63

$$C \text{ is approximately } 3000 \times 11.63 = 34,860 \text{ bps}$$

**Channel capacity C is approximately 34,860 bps (about 34.86 kbps)**

### Question 3

Scenario: A hospital uses wireless patient-monitoring devices that send heart-rate and blood-pressure readings continuously. Because wireless interference can flip a single bit, the hospital uses Hamming (12,8) code so that a single-bit error can be corrected instantly at the receiver, without waiting for retransmission (critical for real-time patient monitoring).

#### Step 1: Determine the number of redundant (parity) bits

For  $m = 8$  data bits, the number of redundant bits  $r$  must satisfy:  $2^r \geq m + r + 1$

$$\text{For } r = 4: 2^4 = 16 \geq 8 + 4 + 1 = 13 \rightarrow \text{condition satisfied}$$

So  $r = 4$  parity bits are needed, giving a total codeword length of  $m + r = 8 + 4 = 12$  bits  $\rightarrow$  Hamming (12,8) code.

#### Step 2: Bit positions in the 12-bit codeword

Parity bits are placed at positions that are powers of 2 (1, 2, 4, 8); data bits occupy the remaining positions.

| Position | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit      | P1 | P2 | D1 | P4 | D2 | D3 | D4 | P8 | D5 | D6 | D7 | D8 |

Each parity bit checks a specific set of positions (based on binary position numbers):

P1  $\rightarrow$  positions 1, 3, 5, 7, 9, 11      P2  $\rightarrow$  positions 2, 3, 6, 7, 10, 11

P4  $\rightarrow$  positions 4, 5, 6, 7, 12      P8  $\rightarrow$  positions 8, 9, 10, 11, 12

#### Step 3: Worked example (encoding)

Assume an 8-bit sample reading  $D1D2D3D4D5D6D7D8 = 10110010$  is to be sent. Using even parity for each parity group:

| Parity | Covers (excl. itself) | XOR of data bits                                                    | Value |
|--------|-----------------------|---------------------------------------------------------------------|-------|
| P1     | 3,5,7,9,11            | $1 \text{ xor } 0 \text{ xor } 1 \text{ xor } 0 \text{ xor } 1 = 1$ | 1     |
| P2     | 3,6,7,10,11           | $1 \text{ xor } 1 \text{ xor } 1 \text{ xor } 0 \text{ xor } 1 = 0$ | 0     |

|    |            |                         |   |
|----|------------|-------------------------|---|
|    |            | = 0                     |   |
| P4 | 5,6,7,12   | 0 xor 1 xor 1 xor 0 = 0 | 0 |
| P8 | 9,10,11,12 | 0 xor 0 xor 1 xor 0 = 1 | 1 |

Transmitted 12-bit codeword: 1 0 1 0 0 1 1 1 0 0 1 0

#### Step 4: Error correction at the receiver

Suppose electromagnetic interference flips bit 6 (D3) during transmission: 1 becomes 0.

Received codeword: 1 0 1 0 0 0 1 1 0 0 1 0

The receiver recomputes each parity check (C1, C2, C4, C8) using even parity:

| Check                        | Result       |
|------------------------------|--------------|
| C1 (positions 1,3,5,7,9,11)  | 0 (matches)  |
| C2 (positions 2,3,6,7,10,11) | 1 (mismatch) |
| C4 (positions 4,5,6,7,12)    | 1 (mismatch) |
| C8 (positions 8,9,10,11,12)  | 0 (matches)  |

Syndrome = C8 C4 C2 C1 = 0 1 1 0 = decimal 6

The syndrome value (6) directly points to the position of the corrupted bit (position 6). The receiver simply flips bit 6 back from 0 to 1, recovering the original codeword - without any retransmission.

Relevance to the hospital scenario: Since Hamming (12,8) can locate and correct any single-bit error instantly using the syndrome, the doctors continue to receive accurate, real-time heart-rate/blood-pressure data even when wireless interference corrupts one bit, avoiding the delay that retransmission (ARQ) would introduce - which is essential for critical patient monitoring.

## Question 4

### a) Subnetting 172.16.10.0/24 with a /27 mask

Default mask /24, new mask /27 -> borrowed bits = 27 - 24 = 3

Number of subnets =  $2^3 = 8$

Host bits remaining = 32 - 27 = 5 -> Hosts per subnet =  $2^5 - 2 = 30$  usable hosts

Block size (increment) = 256 / 8 = 32, so subnets are:

| Subnet # | Network Address | Usable Host Range             | Broadcast Address |
|----------|-----------------|-------------------------------|-------------------|
| 1        | 172.16.10.0     | 172.16.10.1 - 172.16.10.30    | 172.16.10.31      |
| 2        | 172.16.10.32    | 172.16.10.33 - 172.16.10.62   | 172.16.10.63      |
| 3        | 172.16.10.64    | 172.16.10.65 - 172.16.10.94   | 172.16.10.95      |
| 4        | 172.16.10.96    | 172.16.10.97 - 172.16.10.126  | 172.16.10.127     |
| 5        | 172.16.10.128   | 172.16.10.129 - 172.16.10.158 | 172.16.10.159     |
| 6        | 172.16.10.160   | 172.16.10.161 - 172.16.10.190 | 172.16.10.191     |
| 7        | 172.16.10.192   | 172.16.10.193 - 172.16.10.222 | 172.16.10.223     |
| 8        | 172.16.10.224   | 172.16.10.225 - 172.16.10.254 | 172.16.10.255     |

IP 172.16.10.78 lies between 64 and 95, so it belongs to Subnet 3:

**Network address = 172.16.10.64    Broadcast address = 172.16.10.95**

Checking 172.16.10.95: this value lies within the range 172.16.10.64 - 172.16.10.95, so it falls within the same subnet as 172.16.10.78 (in fact it is that subnet's broadcast address).

**b) IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329**

Compression (shortest form): Remove leading zeros in each group, then replace the longest run of consecutive all-zero groups with "::" (used only once).

2001:0db8:0000:0000:0000:ff00:0042:8329    ->    2001:db8::ff00:42:8329

Expansion (back to full notation): Restore the omitted zero groups and pad each group back to 4 hex digits.

2001:db8::ff00:42:8329    ->    2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix with /64: The first 64 bits (first 4 groups) form the network prefix:

Network prefix = 2001:0db8:0000:0000::/64    (i.e. 2001:db8::/64)

Total possible host addresses: remaining bits = 128 - 64 = 64 bits

Number of host addresses =  $2^{64}$  = 18,446,744,073,709,551,616

## **Question 5**

Scenario summary: A server (10 Gbps link) transmits exam data simultaneously to 8,000 students whose access links are much slower (100 Mbps, 20 Mbps, 5-10 Mbps). Slower-end students experience incomplete downloads, repeated retransmissions, delayed ACKs, and a frozen portal, while the server keeps transmitting at high speed even though many receivers cannot keep up - causing receiver buffer overflow and extra retransmissions.

### **Primary networking problem**

This is fundamentally a FLOW CONTROL problem - a mismatch between the sender's transmission rate and the receiver's reception/processing rate. Because the network as a whole is also overloaded by 8,000 simultaneous connections, an element of network CONGESTION also occurs, but the specific symptom described (fast sender overwhelming a slower receiver, causing its buffer to overflow) is the classic signature of a flow-control failure.

### **How the sender-receiver speed mismatch affects communication**

1. Buffer overflow: The receiver has a finite buffer to hold incoming data before the application processes it. When the sender pushes data faster than the receiver (limited by its 5-20 Mbps link) can drain the buffer, the buffer fills up and any further incoming data is dropped.
2. Packet loss and retransmissions: Dropped packets are detected (via timeout or duplicate ACKs) and must be retransmitted, wasting bandwidth that is already scarce for slow-link students.
3. Delayed acknowledgments: Because the receiver is busy/overloaded, ACKs are generated late; the sender, following protocols like TCP, may misinterpret this as loss and retransmit unnecessarily, compounding the problem.
4. Increased network traffic and congestion: The extra retransmissions from thousands of students add to overall network load, pushing the shared network further toward congestion and slowing everyone down, including students who might otherwise have received data smoothly.
5. Freezing of the portal: The repeated cycle of overflow, loss, retransmission and delay causes the perceived "freezing" experienced by students on slower connections.

### **Recommended solution**



This calls for proper flow control mechanisms, such as a sliding-window protocol where the receiver advertises its available buffer space (receive window) so the sender only transmits as much data as the receiver can currently accept. In addition, congestion-control techniques (e.g., staggered/rate-limited delivery, TCP congestion window, or CDN-based load distribution) would prevent the shared network from being overwhelmed when all 8,000 students connect at the same time.